DOMINANCE IS UNCHANGED BY A FIELD EXTENSION OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The lemma is placed in the section on properties of morphisms local on the target, and its conclusion is used there; the card reproduces the statement as delivered, without adding the locality phrasing.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A morphism from the spectrum of a field extension to the spectrum of the base field is universally open. Morphisms, Lemma morphisms-lemma-scheme-over-field-universally-open, invoked at proof lines 5869-5871. Makes the base change of the target open over the target, which is what the preservation statement for dominant morphisms needs.
- Dominant morphisms are preserved by open base change. Morphisms, Lemma morphisms-lemma-open-base-change-dominant, invoked at proof lines 5873-5875. Gives the forward direction.
- The base change of the target along the field extension is surjective onto the target, and dominance is stable under composing with a surjective morphism on the target side. Morphisms, Lemma morphisms-lemma-base-change-surjective, invoked at proof lines 5877-5878. Gives the converse direction, together with the identification of the two composites.

Source reference. Original lemma 01S1: sources/stacks-chow/tags.txt, source lines 2322–2322. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0H8F: sources/stacks-chow/tags.txt, source lines 21490–21490. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let E/k be a field extension. Then a morphism $X \to Y$ over k is dominant if and only if the pullback $X_E \to Y_E$ is dominant.*

Proof. By Morphisms, Lemma 0383, the morphism $\mathrm{Spec}(E) \to \mathrm{Spec}(k)$ is universally open. So $Y_E \to Y$ is open. Therefore, if $X \to Y$ is dominant, Morphisms, Lemma 0H8F gives that $X_E \to Y_E$ is dominant. Conversely, suppose that $X_E \to Y_E$ is dominant. The morphism $Y_E \to Y$ is surjective by Morphisms, Lemma 01S1, and so the composition $X_E \to Y_E \to Y$ is dominant. This is also the composition $X_E \to X \to Y$, and so $X \to Y$ is dominant. □